CS Chapter 6 CRT questions

2. Explain the difference between method declaration and method body.
A method declaration (the method header) sets up the method's name, return type, access modifier, and parameters so Java knows how it can be called. The method body is the actual code inside the curly braces {} that runs whenever the method gets executed.

3. What type of keyword is used to change the access level of a method?
An access modifier (like public or private).

4. What is another word used for describing the access level of a method?
Visibility.

5. Explain the scope of each of the variables in the code below:
- var3: Declared inside method1(), so its scope is the entire method1() body.
- var4: Declared in the for loop header in method1(), so its scope is restricted to just that loop block.
- var1: Declared inside main(), so its scope is the entire main() method body.
- var2: Declared in the for loop header in main(), so its scope is restricted to just that loop block.

6. Write a method declaration for each of the following descriptions:
a) public static int getVowels(String str)
b) public static int extractDigit(int num)
c) public static String insertString(String str, int index)

7. a) How does the compiler distinguish one method from another?
The compiler looks at the method signature, which is the method's name combined with the number, order, and data types of its parameters.
b) Can two methods in the same class have the same name? Explain.
Yes, as long as they have different parameter lists (different number, types, or order of parameters). This is called method overloading.

8. a) What is the return statement used for?
It exits the method and sends a result value back to wherever the method was called.
b) How many values can a return statement send back to the calling statement?
Only one value.

c) How is the declaration of a method returning a value different from the declaration of a method that does not return a value?
A method that returns a value specifies its return type (like int, double, or String) in the header, while a method that doesn't return anything uses the keyword void.

9. Find and explain the error in the code below:
In main(), calling doSomething(); on its own line discards the integer value returned by the method. Additionally, the line num = doSomething(); causes a compiler error if num hasn't been declared with a data type (int num;) beforehand.

11. Determine if each of the following are true or false. If false, explain why.
a) True. Breaking a bigger task down into smaller methods is called procedural abstraction.
b) False. A method call just uses the method name and arguments in parentheses; the declaration is where the method is defined.
c) False. A void method does not return any value.
d) False. An access modifier sets the visibility of a method (like public or private), not its return type.
e) True. The static keyword declares it as a class method.
f) False. Parameters go inside parentheses (), not curly braces {}.
g) False. Local variables are only accessible within the specific block or method where they're declared.
h) True. You can change a parameter variable's value inside the method, but changing primitive types won't affect the original argument outside.
i) False. Method overloading means having multiple methods with the exact same name but different parameter lists in the same class.
j) True. The return statement is used to send a value back to the calling statement.
k) False. A precondition describes what conditions must be true before the method is called, not just the parameter types.
l) False. A postcondition describes the result or state after the method finishes, not how the code achieves it.